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Digital twin technology for thermal comfort and energy efficiency in buildings: A state-of-the-art and future directions

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ABSTRACT

In recent years, the integration of digital technologies has grown rapidly in the field of thermal comfort and energy efficiency for buildings. The concept of a digital twin, incorporating multiple digital technologies, has gained increasing attention. The literature lacks a review of the digital twin concept in thermal comfort and energy consumption for existing buildings. This paper conducts a review of the current state-of-the-art in digital twin (DT) technology for thermal comfort/ energy consumption in buildings. The review employs a scientometric approach and examines various technologies used in creating DTs and a systematic analysis of the methods, technologies, algorithms, and approaches used in digital twin experiments. The results show a growing number of studies in this area, with a focus on thermal comfort monitoring, visualization, tracking, energy management, prediction, and optimization for existing buildings. Furthermore, the prediction of energy consumption using algorithms such as Artificial Neural Networks (ANN), Artificial Intelligence (AI), deep neural networks, and YOLOv4 have been used in buildings. However, the wider adoption of a DT that can facilitate occupants, and thermal sensations, enhance human-centered solutions, and improve energy prediction levels are necessary. There is a need for further international collaboration to expand the studies on digital twins for thermal comfort and energy efficiency. The review highlights the limitations and areas of improvement, such as the limited adoption of sensors for environmental measures, the need for more focus on the subjective perception of occupants, and the need for more comparative studies of algorithms for predicting energy consumption. Further studies can be conducted in areas such as understanding occupant psychological responses/behaviors to comfort in the digital world. This will enhance a more consolidated and robust validation for building performance.

1. Introduction

In the era of the digital age, different concepts and technologies are transforming the built environment in making construction, homes, and cities smart for users' convenience. Digital technologies are tools that have been adopted in enhancing productivity, and performance in a project and built environment [1].These technologies include Augmented Reality/Virtual Reality (AR/VR), Internet of Things (IoT), blockchain, Artificial Intelligence (AI), Mixed Reality (MR), Building Information Modelling (BIM), and Digital Twin (DT). DT as one of the trending technologies can digitally represent physical entities and ensure the bi-directional link between both entities for information, visualization, monitoring, management and so on [2,3]. DT incorporates several technologies such as BIM and IoT to create digital entities and transfer information from the physical building to the digital model respectively. DT has various capabilities depending on the technologies integrated and what DT will be used to perform.

DT has been applied in several areas of sectors such as manufacturing, transporting, health, aviation, and agricultural sectors but its application in the built environment is at its nascent stage [4]. In the construction industry, DT has been adopted in the design stage serving as BIM i.e., a digital model having geometrical and contextual information [5–8] This has also been adopted in the construction stage for training workers on how to ameliorate musculoskeletal injuries, structural integrity, and site logistics [9–11]. In the built asset, there are also some applications of DT ranging from asset monitoring, fault/anomaly detection of systems, thermal comfort, and energy management [12–14]. However, the thermal comfort of occupants in the building is one of the critical factors that need to be considered in improving the productivity of well-being and performance/activity of occupants. Thermal comfort acceptance varies with different kinds of people depending on their activities, state of mind, personal and environmental factors.

Different thermal comfort models are used for predicting various kinds of contexts and settings for the preferences of people in different locations [15–17]. There is no perfect model or consensus on one model for predicting the thermal comfort of users because of several factors such as climate, psychological, and physiological characteristics of each individual. This has led to an adaptive and personal/individual

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Nomenclature

Abbreviation

DT	digital twin
IoT	Internet of Things
AR/VR	augmented reality/virtual reality
XR	extended reality
MR	mixed reality
BIM	building information modelling
AI	artificial intelligence
PLM	product life management
SVM	support vector machine
PCA	principal component analysis
ANN	artificial neural network
KNN	K-neural neighbors
GNN	graph neural network
YOLOv4	you only look once
GA	genetic algorithms
GP	gaussian process
GNN	graph neural network
Build2Vec	building representation in vector space
HVAC	heating, ventilation, and air conditioning
O&M	operation and maintenance
FM	facility management
MEP	mechanical, electrical and plumbing
PRISMA	preferred reporting items for systematic reviews and meta-analysis
SCADA	supervisory control and data acquisition
BEMS	building energy management system
PMV	predicted mean vote
PPD	predicted percentage of dissatisfied
CBE	center for built environment
TSV	thermal sensation votes
NZEB	net-zero energy building
MPC	model predictive control
ANSYS	analysis of systems
TRNSYS	transient system simulation tool
E+	energy plus
ASHRAE	American society of heating, refrigerating and air-conditioning engineers
MATLAB	matrix laboratory
DNN	deep neural networks

comfort model which has superseded the rational model [18,19]. Understanding the personal thermal comfort of users will enhance energy management, minimize consumption of energy and comprehend implications of actions while maximizing thermal comfort. A review was conducted to distinguish the setting and context of digital twin applications for energy efficiency in buildings. The results enabled the classification of previous studies into optimization design, occupant comfort, building operation and maintenance, and energy consumption simulation [20]. Other reviews conducted include the application of digital twins [21–24], drivers of DT [25]; challenges [22,23,26,27]; benefits [23] in the construction industry. However, there is less focus on current research on DT applications for thermal comfort and energy.

This research aims to answer the following Research Questions:

1 What is the current scope of research on digital twins for thermal comfort and energy?
2 What are the tools, technologies, and software adopted for creating digital twins for thermal comfort and energy?
3 What are the other conceptual or applications that need to be tapped?

This paper reviews digital twins for thermal comfort and energy consumption in existing buildings. It discusses the tools, technologies, software adopted, and further research areas which contribute to the existing body of knowledge. These objectives will be achieved by the following approach/method; reviewing existing studies on DT for thermal comfort and energy consumption, carrying out an analysis of tools, and technologies adopted for monitoring and visualization of thermal comfort, providing an in-depth discussion on the different methods adopted, comprising gaps to be filled and areas of further studies

The remaining part of the review paper is as follows; Section 2 gives an overview of the digital twin, and its application in various stages of the lifecycle of project and asset while Section 3 discussed the methodology adopted in this review paper for bibliometric and systematic. Section 4 elucidates the scientometric analysis result, while Section 5 discusses tools, technologies, and software adopted for thermal comfort and energy consumption in existing buildings. Discussions, limitations, and future directions of research were elaborated in Section 6 while the conclusion was provided in Section 7

2. Literature review

2.1. Overview of digital twin

Digital twin contains three elements which are the physical space, Digital or virtual space, and connected data (bi-directional exchange). The Digital twin for Product Life Management (PLM) means the two-system (digital and physical) are linked together throughout the lifecycle of the system. In 2011, the concept was expanded and was refined as a “Digital twin” and got more attention in 2015 and 2017. It was adopted by astronautics and aerospace and NASA used it for technology roadmaps and sustainable space exploration This DT idea was proposed for next-generation fighter aircraft [28,29]. A digital twin is the full representation of physical entities in digital space as shown in Fig. 1. According to [30], digital twins can be categorized into three types; Digital twin prototype (DTP), Digital twin instance (DTI), and Digital twin aggregate (DTA). DTP is constructed as a digital model of an object that has not been produced/build in the physical world, DTI is a digital twin itself while DTA is the combination of several virtual twins

2.2. Application areas of digital twin

DT has been prominent in various sectors such as manufacturing, aviation, health, and transportation. In the manufacturing sector, Rosen, von Wichert [32] examined the four aspects that are important for future manufacturing which are modularity, connectivity, autonomy, and DT. Schleich, Anwer [33] discussed how digital twins can shape design and production engineering and found that DT can enhance product redesign, scalability, interoperability, and expansibility. Uhlemann, Schock [34] emphasized that DT allows real-time production control in the industry. Grieves and Vickers [30] looked at how DT can mitigate unpredictable behavior in a complex system. DT was applied in the design, manufacturing, and service of the whole product lifecycle which revealed that it brings efficiency, smartness, and sustainable product life cycle management [35]. This means DT can be employed in lifecycle management, in which the benefits can cut across the design, construction, operation, and maintenance phases of a built asset.

In the construction sector, several areas where digital twins can be applied are lifecycle management, big data, real-time monitoring, decision-making, or performance prediction. Opoku, Perera [24] reviewed six areas of application in the lifecycle of a project which are BIM, structural system integrity, facilities management, monitoring, logistics processes, and energy simulation. DT can reduce construction and operating costs, increase productivity and collaboration, improve safety, and optimize asset performance and sustainability. Opoku, Perera [24] also noted that most of the digital twin applications focused on a single lifecycle phase. Most studies in construction have focused more on the design engineering, operation, and maintenance

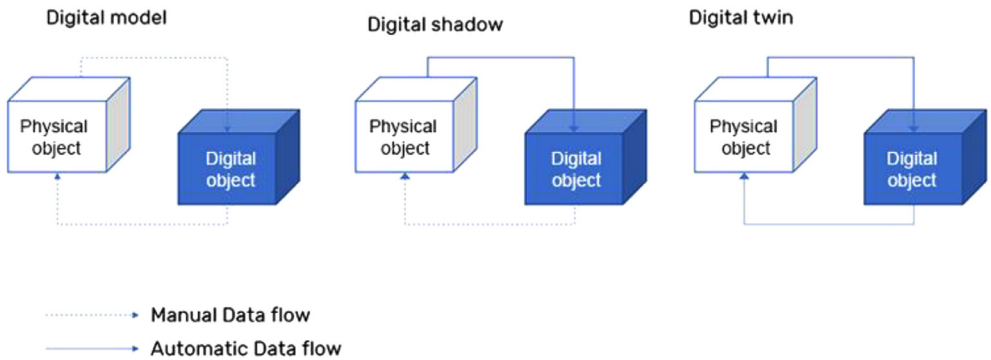

Fig. 1. Pictorial representation of Digital model, Digital shadow, and Digital twin Adapted from [31].

Table 1
Summary of a few applications of Digital twin for different phases of the project.

Phases	Applications	Refs.
Design and engineering phase	Building information modeling (Net zero building, sustainable railway, existing building, optimization, cost estimation)	[5,6,8,38,39]
Construction phase	Structural system integrity	[9]
	Construction site logistics	[11]
	Ameliorating musculoskeletal injuries for construction workers	[10]
	Monitoring and management, quality and safety monitoring, machine & material monitoring	[40–42]
Operation and maintenance phase	Asset/Facilities management	[12]
	Anomaly Detection and Asset Monitoring	[8,43]
	Energy simulation, usage, and management	[13,44-47]
	Indoor environmental monitoring and Thermal comfort	[14,48-52]
Demolition phase	None applicable	N/A

phases, with few papers on the construction phase neglecting the demolition/refurbishment stage. This also corresponds to the research conducted by Kan and Anumba [36] on application and various research opportunities in DT.

Errandonea, Beltrán [37] classified the application of DT into five groups these are design, process/logistic/production, prognostic health monitoring, maintenance, and life cycle in general. However, prognostic health monitoring and maintenance can be combined into the operation and maintenance phase of systems or projects while process/logistic/production is in line with the construction phase for infrastructure. The lifecycle involves the four stages of application design, construction, operation and maintenance, and demolition stage as revealed in Table 1.

2.3. Digital twin in the built assets

Built assets necessitate planning, design, management, maintenance, and monitoring of functional and physical characteristics of the infrastructure [12]. Asset management is one of the areas where attention has been focused on transforming existing document-oriented asset management practices into model-based, data-driven, automated processes of asset management. This can be executed using various technologies such as big data, AI, cloud computing, Virtual reality, AR, IoT, and BIM. The digital twin has recently been one of the tools or technology used in asset management for buildings [53]. DT was used to examine the anomaly detection and diagnosis of the subsystem [8].

There are three types of models adopted for the built asset which are quantitative model-based, qualitative, and history-based. According to Kallab, Chbeir [54], the quantitative-based model uses mathematical and physical models which are called white-box models. Qualitative models userule-based and if-then-else rules. Operational values like pressure, flow, and temperature are kept within pre-set values or set points. The history-based methods are referred to as data-driven methods (black box models). Historical data from the building management system is used to estimate trends, patterns, and outliers. Machine learning methods such as SVM, PCA, and ANN are applied to fault detection in components where sensors are adopted [55].

2.3.1. Digital twin and operations/facility management

DT is gaining wide adoption in the operations, and facility/asset management of the building. Xie, Lu [12] and Lu, Xie [8] developed a digital-enabled framework for asset condition monitoring and anomaly detection. A Bayesian change point methodology was used to reveal anomalies in real time for HVAC. This was performed on the heating and cooling pump of the HVAC system which was carried out at the Centre for Digital Built Britain (CDBB) west Cambridge digital twin pilot. The two sub-tasks of the framework detection were to implement Bayesian change detection to find out symptoms and to distinguish between real anomalies and Normal operations. Peng [56] proposed a continuous lifecycle integration framework for a large hospital in China. A DT software system with real-time visual management and AI diagnosis was created and used in a newly built DT control center.

Zhao [57] reviewed four illustrative case studies where DT was used in the Operation and Maintenance (O&M) phase. The selection of four case studies was based on three criteria which are the availability of data, different case characteristics, and useful variations of the dimensions of theoretical interest. The findings were used to propose a conceptual framework that could be used as the catalyst for improving the efficiency of FM. The study emphasized that further research should examine how these technologies can tackle environmental issues, energy consumption, and carbon emissions. Cheng [55] proposed a data-driven framework for FM of indoor systems. This approach adopted ANN and SVM to predict the future conditions of MEP components which is based on condition monitoring of equipment which has helped with the scheduling of maintenance tasks.

3. Methodology

This review uses a mixed method approach that combines scientometric analysis and critical review to understand the current scope

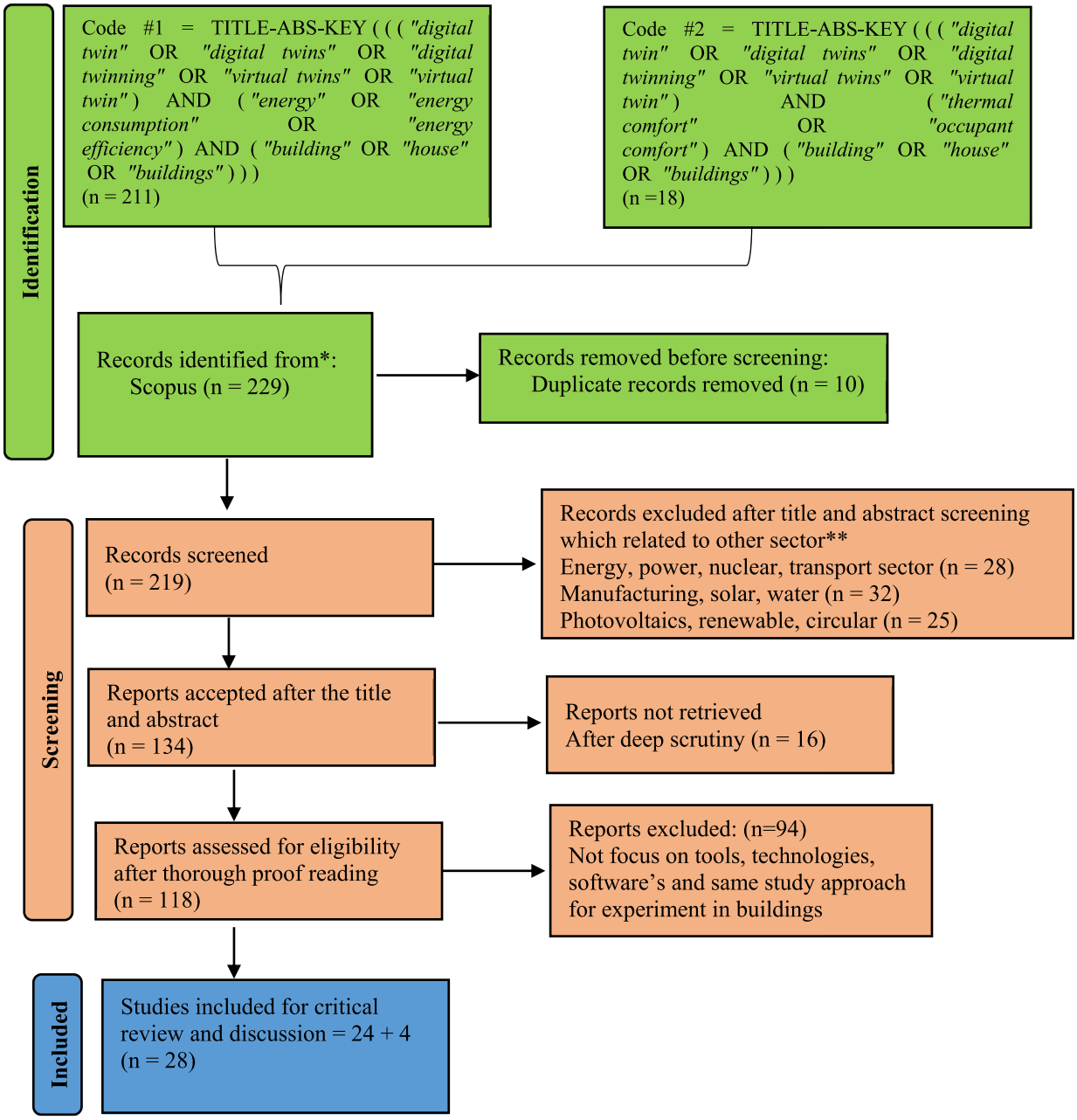

Fig. 2. PRISMA flow diagram of paper review.

of Digital Twins (DT) in thermal comfort and energy consumption in buildings. A scientometric analysis specifically science mapping is used to visualize important patterns and trends in a large body of literature [58,59]. Data was collected from the Scopus database using specific keywords and a 10-year time frame. Duplicates were removed and the remaining 229 papers were imported to Endnote and analyzed using the VOS viewer software. The analysis includes keyword co-occurrence, authorship, countries, and sources to understand the relationship of the network and identify gaps in the field. Additionally, a critical review is conducted to provide a qualitative understanding of the topic. The overview is shown in Fig. 2 [60].

3.1. Selection of database and data collection

This review used Scopus as the main database for data collection due to its broad coverage of research publications related to construction management [61] and the availability of more bibliometric data compared to other databases such as Google Scholar and Web of Science. Other databases were not considered as they were not as commonly used in construction-related research [62,63].

3.2. Keywords selection strategy, yardstick for inclusion and exclusion

The review used Scopus to extract bibliographic data using selected keywords based on the research scope. Scopus was mainly used because using the keywords it has many papers compared to the web of science. Also, most of the papers on the web of science are in Scopus after a critical check in the database search. A desktop review of relevant papers was conducted to test the reliability of the keywords before searching in the database. The first search string resulted in 211 papers after applying inclusion and exclusion criteria. The second search string resulted in 18 papers. The criteria for inclusion and exclusion included papers from environmental science, engineering, computer science, and energy, published in the last 10 years from 2012 to October 2022 and written in

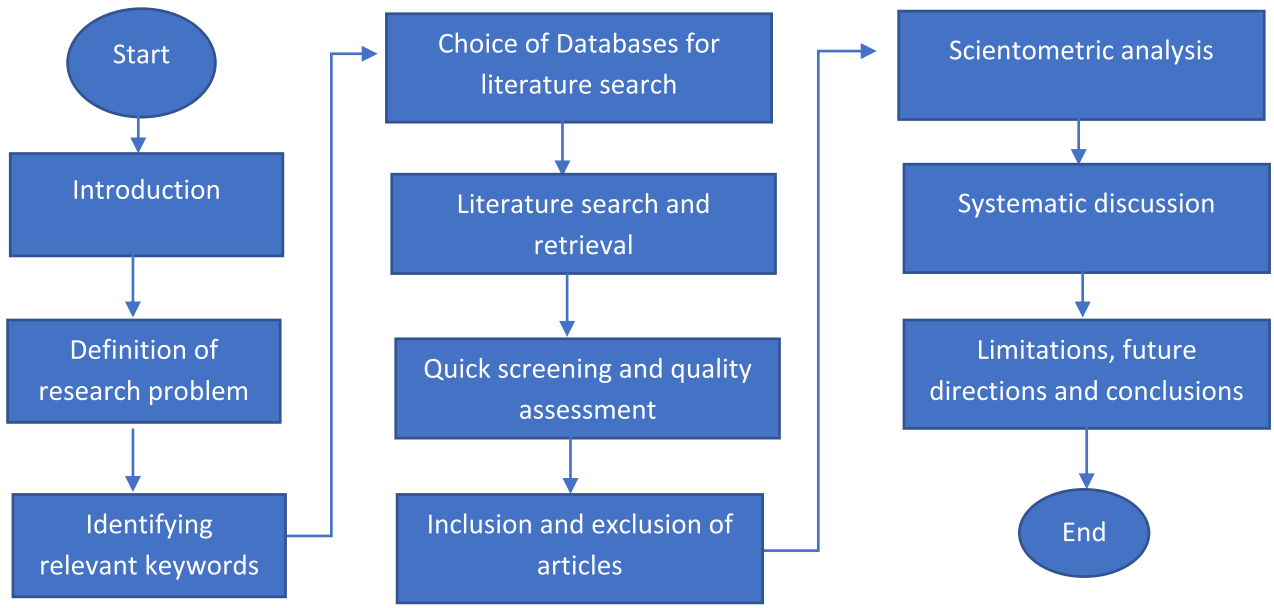

Fig. 3. Overview of the paper.

English. A total of 229 papers were found in the database and imported to reference managers.

3.3. Selection of software tools for analysis and critical analysis review

A scientometric analysis using the Scopus database and VOS viewer software was used to map the structure, patterns, and thematic clustering of the scientific domain [58]. The bibliographic data was filtered and scrutinized to remove duplicates, and the remaining papers were used to create keyword co-occurrence, authorship, countries, and source map to show the relationship of the network. The co-occurrence of keywords helps to understand the links to the topic used by previous researchers in the field and identify gaps in the field. The clustering of keywords in VOS viewers helps researchers to understand the mainstream of research and where further review is needed. The data was also used to create networks of journal citations, bibliographic coupling, document citation, co-authorship, and country citations, showing the relationships between different studies. Among the 118 papers used for scientometrics, the papers were further critically reviewed to see the methodological process, tools/technologies, and software adopted. However, only one was picked for similar studies and approach

3.4. Steps adopted in the systematic review

The section discusses the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) flow of the paper review and the framework of the whole study. Fig. 2 shows the process of extraction of files from the database, and how duplicates and irrelevant were expunged from the records identified or gathered. Fig. 3 also shows the general overview of the paper

4. Scientometric analysis results

4.1. Publications per year

After reviewing 229 papers, 118 relevant articles were extracted and analyzed. The majority were journal articles (84) and a smaller number were conference papers (34). The number of publications per year was analyzed and found to have slow growth in 2018 and 2019, followed by a rapid increase in 2020 during the Covid-19 pandemic as shown in Fig. 4. The research focus shifted from integrating DT with SCADA systems and visualized 3D models to fault diagnosis, real-time monitoring, machine learning, energy management, interoperability, ontologies, Net zero, AI and deep learning, predictive maintenance, and XR technologies. The trend is expected to continue with an increase in publications in 2022. This suggests that digital twin technology in thermal comfort and energy consumption in buildings is gaining attention from researchers and has significant potential in this field.

In 2020, the research focus was on smart cities and sustainability concerning DT for the built environment [46,64,65]. There was also attention on building energy simulation and performance while in the mid of the same year, researchers examined the optimization of intelligent building, and building energy models as revealed in Fig. 5 [66]. 2021 research trends move toward artificial intelligence for energy efficiency in smart buildings [67,68]. Toward the mid of that year, Bayesian methods and optimization were adopted for building systems [69]. Meanwhile, in 2022, research trends shifted to neural networks such as deep learning, and ANN for energy efficiency and personal comfort in buildings [70–72]. More so, attention is going toward XR technologies for intelligent buildings to improve thermal comfort and energy [73–75].

4.2. Publication per country

In terms of country, the top countries for research on digital twin technology for thermal comfort and energy consumption in existing buildings are the USA and the UK, based on the criteria of having at least 2 publications with 15 citations. The USA had the most publications (16) and citations (307), while the UK had 7 publications and 109 citations as shown in Fig. 6. Only US has a high level of energy consumption which was ranked top two while both have political support and have examined digital twin technology primarily for energy management (IEA, 2021). The research in that context focused on a simulation model for optimization, energy model and management, energy efficiency, energy footprint, modeling and control, and asset and facility management. Few studies in the US have also examined occupant comfort by leveraging DT. China, Italy, and Australia also had a high number of publications, but fewer citations. The articles focusing on China cover DT for smart and sustainable buildings and cities. This is often related to information, energy efficiency control, optimization, and consumption, energy retrofit, prefab, and lifecycle integration. However, studies weren't conducted in relation to the thermal comfort of occupants. For the US, this is also likely similar to China, studies were also DT for

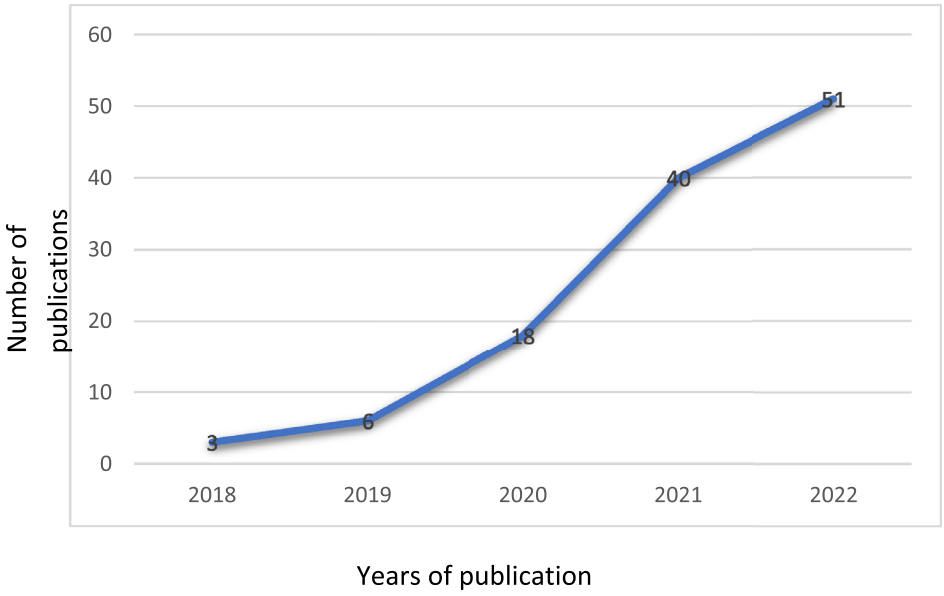

Fig. 4. Publication per year.

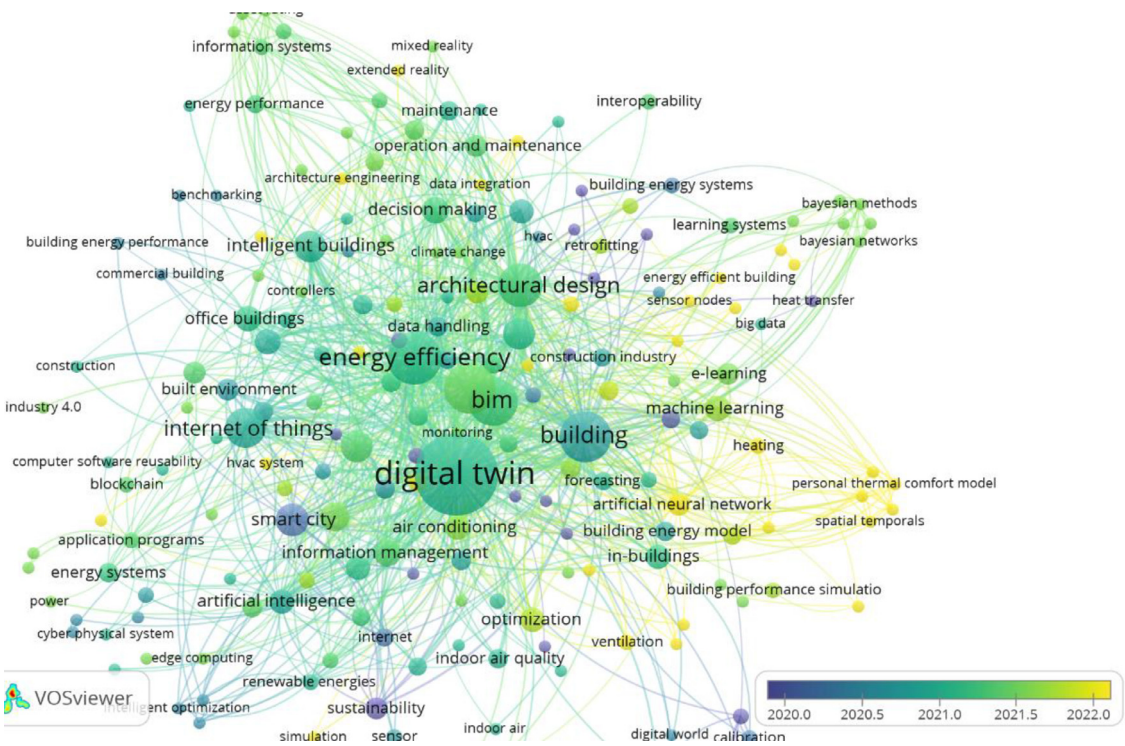

Fig. 5. Overlay visualization map of keywords occurrence.

sustainable smart buildings and cities. Other countries with notable research in this field include Finland, Spain, Switzerland, Singapore, and Canada. However, some top energy-consuming countries, such as India, Russia, Japan, Brazil, and South Korea, were not represented in the research. This suggests that further research is needed in these countries, as well as in Africa and South America.

4.3. Publication per document source

The research on digital twin technology for thermal comfort and energy consumption in existing buildings was found in various journals and conferences. 13 sources met the criteria of having at least 1 document with 20 citations. The *Journal of Management in Engineering (JME)* was the top source with 3 documents and 160 citations which is shown in Table 2. These papers mainly discussed using digital twin technology for risk-informed decisions in cities. *Sustainability,* the second-highest source with 7 documents and 142 citations, focused on BIM and IoT sensor integration for indoor condition and energy consumption, DT for sustainable assessment and sustainable comfort monitoring, AI-based sustainable smart energy city development, simulation modeling, and NetZero building. The Journal of *Energy and Buildings,* the third-highest source with 7 documents and 77 citations, these papers discussed Data driven DT for indoor lighting with aid of BIM and computer vision, a Simulation model of the building system, GIS data for energy mod-

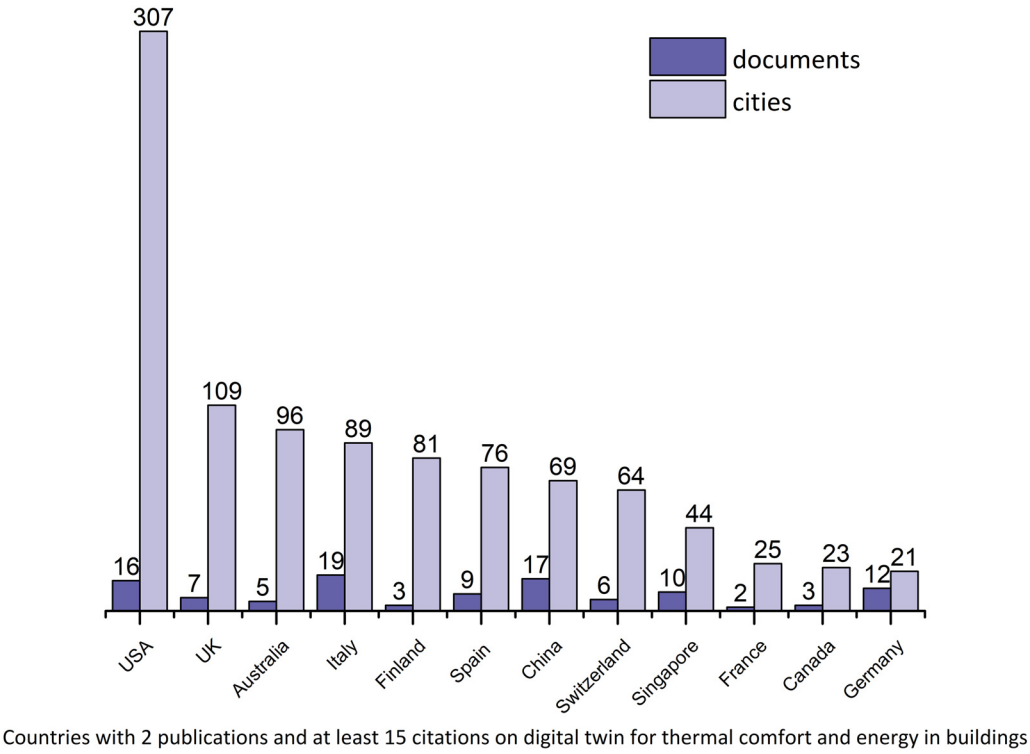

Fig. 6. Countries of publication.

Table 2
Number of publications per source.

Source	Documents	Citations	Total link strength
Journal of Management in Engineering	3	160	1
Sustainability	7	142	7
Energy and Buildings	7	77	6
9th International Conference on Intelligent Systems 2018: Theory, Research, and Innovation in Applications, 2018 – Proceedings	1	63	1
Buildings	6	50	1
Journal of Building Engineering	1	49	2
Applied Sciences	2	43	3
Journal of Information Technology in Construction	1	43	3
Energies	7	34	10
Advances in Civil Engineering	3	31	3
Building and Environment	4	29	1
Sustainable Cities and Society	3	21	1
Engineering, Construction, and Architectural Management	1	20	1

elling and retrofit, and Bayesian optimization of the model for HVAC. Other notable sources included the *9th International Conference on Intelligent Systems* based on smart city platform enabling DT, the Journal of *Buildings* (discussing mainly carbon emission benchmarking system and performance, DT framework for energy savings, DT predictive maintenance i.e., fault diagnosis and detection, DT for indoor air quality based on BIM, BIM-based tool kit for renovation, Differentiation on Digital model, shadow and DT, DT conceptualization for circular construction, DT for Machine learning approach regards operation and maintenance, and adoption of blockchain through DT), and the *Journal of Building Engineering* (based on a systematic review on applications of DT in the construction industry). In terms of link strength, the journal with the strongest link was Energies followed by Sustainability. This suggests that these journals have the most connections and collaborations between papers in the field of digital twin technology for thermal comfort and energy consumption in existing buildings.

4.4. Most cited publications

This review analyzed the most cited publications on digital twin technology for thermal comfort and energy consumption in existing buildings. Out of 118 papers, 21 papers met the criteria of having 15 citations or more. Francisco et al. [64], the most cited author with 75 citations, emphasized the importance of real-time monitoring for digital twin technology implementation. Followed by Kaewunruen et al. [6], 65 citations, discussing how existing buildings can better function and be more sustainable achieving net zero using the BIM platform. This paper was also cited based on the cases examined for photovoltaic and wind systems in achieving Net Zero. Ruohomaki [76], 63 citations, refers to the urban digital twin application or platform that was developed in Helsinki, Finland displaying energy consumption between similar buildings. Table 3 shows other authors with high citations.

Table 3
Top cited publications.

Document	Citations	Method	Area of Focus
Francisco, Mohammadi [64]	75	Case study	Smart meter for Building
Kaewunruen, Rungskunroch [6]	65	Case study	Net Zero for building
Ruohomaki, Airaksinen [76]	63	Case study	Energy in buildings
Opoku, Perera [24]	49	Review	Construction industry
Sepasgozar [77]	46	Review	Built environment
Austin, Delgoshaei [46]	45	Experimental	Residential building
Deng, Menassa [4]	43	Review	Smart building
Ham and Kim [65]	40	Case study	Risk Management
Lydon, Caranovic [78]	35	Experimental	Heating and cooling system
Porsani, de Lersundi [79]	26	Case study	Residential and warehouse building
Zaballos, Briones [52]	26	Case study	Educational building
Tagliabue, Cecconi [80]	24	Case study	Sustainability evaluation
Agostinelli, Cumo [81]	22	Case study	Residential district in Rome
Chong, Gu [82]	20	Review	Energy
Peng, Zhang [56]	20	Case study	Hospital Equipment
Xie, Lu [12]	20	Experiment	Anomaly of thermal comfort
Shahinmoghadam, Natephra [83]	19	Case study	Thermal comfort
Alanne and Sierla [84]	18	Review	Smart Buildings
Coupry, Noblecourt [74]	17	Review	Maintenance in Smart buildings
Al-Ali, Gupta [85]	16	Review	Environmental
Gutiérrez González, Fernandez Bandera [66]	15	Experiment	Energy in building

Table 4
Number of publications per author.

Authors	Affiliation	Documents	Citations	Total link strength
Kaewunruen S.	School of Engineering, University of Birmingham, Birmingham, UK	2	76	0
Chong A.	Department of Building, School of Design and Environment, National University of Singapore	5	33	6
Bandera C.F.	School of Architecture, Spain	2	30	0
Agostinelli S.	Faculty of Architecture, Sapienza University of Rome, Italy	3	28	7
Cumo F.	CITERA Research centre, Italy	4	28	7
Guidi G.	National Agency for New Technologies, Energy and Sustainable Economic Development, Rome, Italy	2	28	6
Tomazzoli C.	Computer Science Department, University of Verona, Verona, Italy	2	28	6
Peng Y.	College of Architecture and Urban Planning, Chongqing Jiaotong University, Chongqing, China	2	22	0
Lu Q.	Bartlett School of Construction and Project Management, Univ. College London, London, UK	2	20	0
Miller C.	Department of the Built Environment, College of Design and Engineering, National University of Singapore (NUS), Singapore	3	13	5
Nielsen H.K.	University of Agder, Grimstad, Norway	2	13	4
Svennevig P.R.	University of Agder, Grimstad, Norway	2	13	4
Svidt K.	Aalborg University, Aalborg, Denmark	2	13	4

4.5. Publication per author and co-authorship network

Out of the 118 papers analyzed, 13 authors met the criteria of having 2 documents with 10 citations. Kaewunruen was the top author with 2 papers and 76 citations, primarily focused on retrofitting existing buildings for net-zero energy using photovoltaic and wind systems (renewable energy) with BIM. Chong was the second-highest author with 5 papers and 33 citations, primarily focused on personal thermal comfort using digital twin technology extracted spatial temporal proximity data with wearable devices. Bandera had 2 papers with 30 citations, focusing on interoperability between BIM and BEM. Agostinelli had 3 papers and 28 citations, primarily focusing on digital twin technology for energy management and renewable energy using AI. Cumo and Guidi had 4 and 2 papers respectively, also focusing on energy management and renewable energy. Tomazzoli and Peng had 2 papers each, with 28 and 22 citations respectively. Other notable authors are listed in Table 4.

4.6. Research focus based on co-occurring keywords

When creating a co-occurrence map for the bibliographic data, it is important to set a minimum number of co-occurrences for keywords to be extracted to properly cluster keywords into themes. Thesaurus was used to remove duplicate keywords which were uploaded as a file in Vos viewer. After removing duplicates, 1130 keywords were analyzed, but only 199 met the criteria of having at least 2 occurrences leading to 10 clusters as shown in Fig. 7 :

Cluster 1 had 36 items and a few keywords are digital twin, artificial intelligence, building energy management, ontology, residential districts, simulation, smart city, sustainability, intelligent optimization, and so on. This can be represented as the "*Optimization of smart buildings*" as shown in Fig. 8. Research efforts have been on machine learning, AI, and optimization. This approach entails achieving intelligent optimization and automation system for energy management [71,81]. Digital twins, AI, and energy management have been prominent areas in the cluster based on node size. Therefore, further studies on how DT can enhance decarbonization and NetZero. *Cluster 2* had 27 items which includes data visualization, fault detection, interoperability, intelligence, model predictive control, real-time, semantics, building automation system, 3D modeling, and so on. This can be called *"predictive maintenance of the intelligent system"*. Based on the studies conducted, attention has been placed on HVAC pumps [8,86]. This can be extended to other systems in the buildings such as gas and water

Cluster 3 had 25 items which are ANN, Bayesian method, case studies, building systems, deep learning, numerical model, machine learning, sensor nodes, computer simulation, and others. This can be connoted as a *"Neural network for building systems"*. This research theme

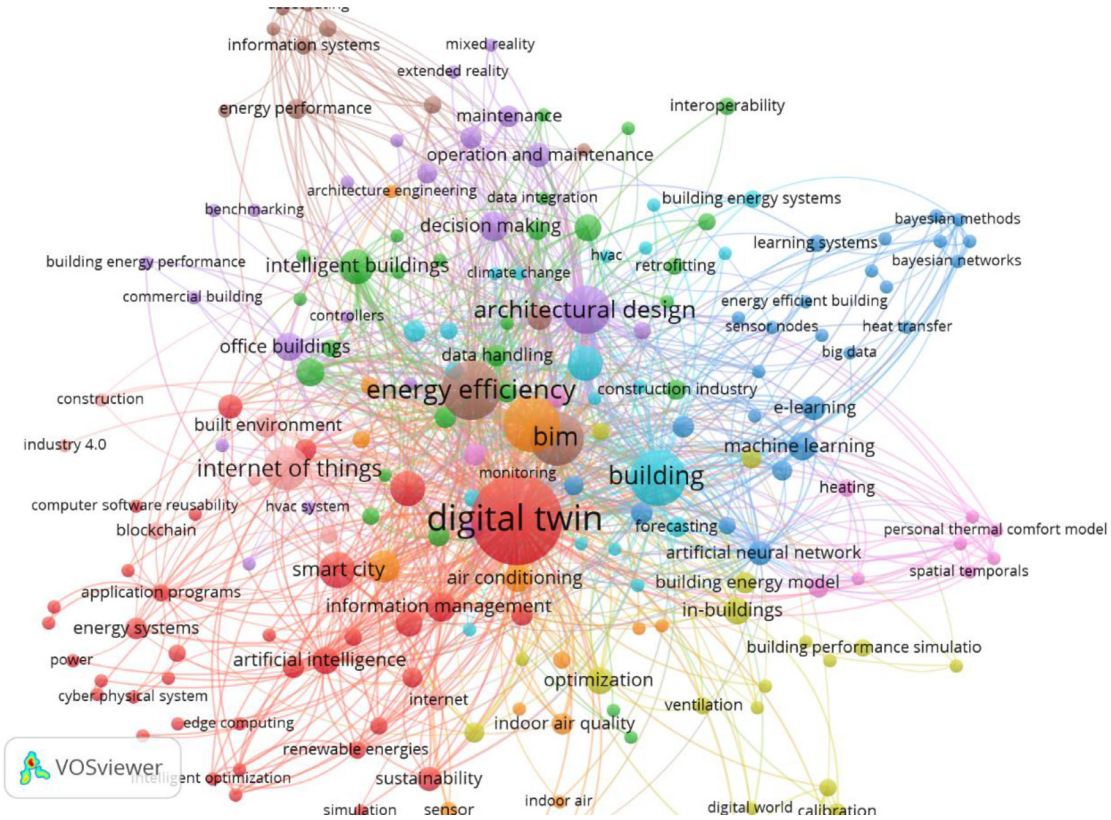

Fig. 7. Network Visualization map of keywords occurrences.

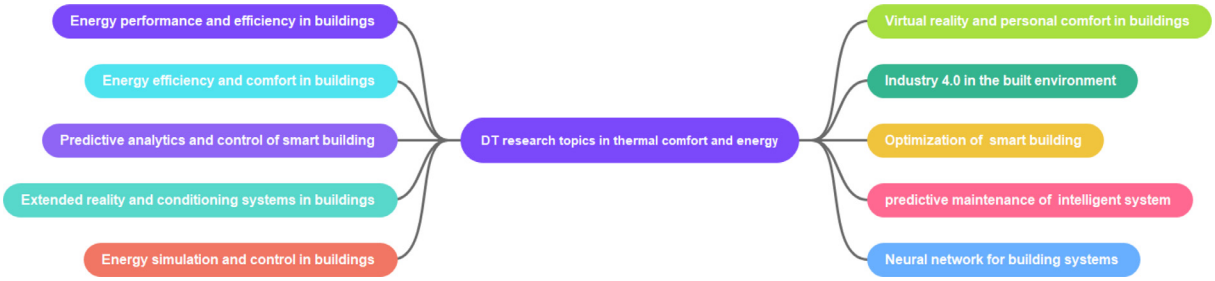

Fig. 8. Framework for major research areas on DT for thermal comfort and energy.

explains how deep neural networks are adopted. The most prominent algorithms are ANN, the Bayesian method, and deep learning which has been examined for building systems [72]. This has helped in understanding the prediction and optimization of systems. *Cluster 4* had 22 items, the prominent keywords for this cluster are building control, energy models, energy simulation, sensors, heritage buildings, and others. This can be represented with *"Energy simulation and control in buildings"*. Here studies have been done in integrating BIM models and BEM models in understanding energy consumption in buildings. Simulation tools coupled with BEM are used in energy reduction and decarbonization [79]. This has also played a significant role in different applications such as retrofit, architectural designs, and optimization

Cluster 5 has 22 items which include Architectural design, AR, Cloud-based, data analytics, decision-making, conditioning systems extended reality, Mixed reality, HVAC systems, buildings, maintenance, and so on. This can be called *"Extended reality and conditioning systems"* in buildings". Research efforts have been carried out on how XR technologies can be explored for maintenance and through cloud platforms for thermal conditions [73–75]. However, interactions and scenarios are still at a nascent stage and are sparsely explored based on hardware and software limitations. Future studies are needed in this space. *Cluster 6* have 22 items which are building energy systems, data-driven model, predictive control, reinforcement learning, real-time system, climate change, smart building, and others. This can be encapsulated with *"Predictive analytics and control in smart building"*. This is predicting and effective control of energy. Other ML algorithms are needed such as Gaussian and SVM

Cluster 7 has 16 keywords including AC, energy efficiency, energy consumption, indoor air quality, quality control, occupant comfort, and so on. This can be ascribed to *"Energy efficiency and comfort in buildings"*. Efforts have been made in using BIM and IoT in visualization and tracking to understand the comfort of people in the environments for different controls and their implications for energy [20,52]. *Cluster 8* has 12 which are BIM, digital twin, energy performance, information systems, operational rating, energy efficiency, and others. This can be called *"Energy performance and efficiency in buildings"*. The energy performance of buildings is not in efficiency but also NetZero and positive energy. The digital twin has been said to help understand energy per-

formance by getting real-time information about assets and operations in the building [6,87]

Cluster 9 has 10 items which are data set, heating, personal thermal comfort, prediction, spatial temporals, Spatio-temporal analysis, thermal comfort model, thermal preferences, and virtual reality. This is called *"Virtual reality and personal comfort in buildings"*. Thermal comfort studies were recently conducted in relation to BIM and VR technologies creating an interface for users to interact [83]. Also, spatio-temporal data from build2Vec was conducted in understanding personal comfort [88]. More studies using other ML models are needed in this area. *Cluster 10* has 9 items which are blockchain, BIM, built environment, internet of things, monitoring, construction, built environment, and industry 4.0. This is connoted by *"Industry 4.0 in the built environment"*. This research theme explains how digital technologies are aiding the built environment for better monitoring, visualization, and performance. Similarly, limited studies have been conducted on how blockchain can be integrated into digital twins based on a conceptual framework [89]. More studies are needed especially in experimental works to enhance trust, transparency, and confidentiality of data and more adoption

5. Systematic discussion

This section critically reviews a selection of published papers on digital twins for thermal comfort and energy consumption in buildings. Initially, 24 papers were chosen that met the criteria outlined in Fig. 2. However, four additional papers published in January 2023 were included as they offered a unique perspective not covered in the other studies making it 28 papers altogether. For thermal comfort aspect, 9 papers were screened out of 18 papers while 85 papers were screened out of 100 papers related to energy. This section examines the methods, technologies, tools, algorithms, and experiments used in the research articles and discusses the findings from case studies and experiments. This helps to identify gaps and areas that have yet to be fully explored for further research.

5.1. Digital twins and thermal comfort in buildings

Thermal comfort refers to the condition of being physically and psychologically comfortable with the surrounding temperature. It is measured by various indices such as Predicted Mean Vote (PMV) and Predicted Percentage of Dissatisfied (PPD) which are the most prominent conventional methods [17]. Thermal comfort is applied in buildings by designing and controlling the heating, ventilation, and air conditioning (HVAC) systems to maintain an optimal indoor temperature, relative humidity, and air movement.

Adaptations and considerations for thermal comfort in building design vary based on the climatic conditions of a region and the cultural preferences of its population. For example; in hot climates, shading, ventilation, and air conditioning may be prioritized to reduce heat stress and maintain thermal comfort. In cold climates, insulation, passive solar heating, and efficient heating systems may be prioritized to maintain warmth. In humid climates, moisture management and ventilation may be prioritized to reduce the effects of high humidity. Cultural preferences for indoor temperature and humidity levels can also greatly vary and building design may take these into consideration. [90]. The design criteria for thermal comfort in buildings can vary based on the climate, use of the building, and the occupants' needs. Building codes and standards, such as ASHRAE 55, provide guidelines for thermal comfort in commercial and residential buildings. These indices use temperature, humidity, air velocity, clothing insulation, and metabolic rate to determine thermal comfort levels [91]. Other factors include individual perception, task being performed, time of day and season

All of these factors play a role in determining thermal comfort levels and should be taken into consideration when designing and maintaining indoor environments. Thermal comfort in indoor environments can be achieved through various measures which includes control of air temperature (Maintaining a comfortable temperature range, typically between 20 and 26 °C with ranges between 14 and 35 °C). Relative humidity control which ranges between 30 and 60% helps to reduce the perception of dry or stuffy air [92]. Air velocity control to ensure that air movement is not excessive and is evenly distributed to avoid drafts and discomfort. Use of insulation is to prevent heat loss and improve thermal comfort, building envelopes and windows should be well-insulated. Radiant temperature control such as Direct sun or cold surfaces can cause discomfort. Shading, blinds, or radiant barriers can be used to reduce this discomfort. Use of natural ventilation by opening windows or vents. Use of HVAC systems to control temperature and humidity levels. Lastly, the use of personal comfort control in adjusting their own temperature such as fans, space heaters, or air conditioning units can improve comfort levels [90].

In building design, it is important to consider individual variability in thermal comfort preferences. Factors such as age, population, and gender can all play a role in determining an individual's preferred indoor temperature and humidity levels [93–97]. Younger individuals tend to prefer warmer temperatures, while older individuals tend to prefer cooler temperatures [98]. Cultural and ethnic background can influence thermal comfort preferences, as different populations may have different norms and expectations for indoor temperature and humidity levels. Some research suggests that women may have slightly different thermal comfort preferences than men.

By taking these differences into account, building design can help to create environments that are more comfortable and welcoming for all occupants, regardless of age, population, or gender and still reduce energy consumption. This is crucial in ensuring that the indoor environment is optimized for thermal comfort and energy efficiency [99,100]. The objective of thermal comfort in building design is to create an environment that is comfortable for occupants by considering factors such as feeling warm or cold, sweating or shivering, tingling or numbness of the skin, perception of air temperature, air velocity, and humidity, discomfort caused by radiant temperature, and discomfort caused by changes in temperature. This combination of factors determines an individual's level of thermal comfort.

Before the advent of digital technologies, various tools such as CBE, thermometers, hygrometers, and anemometers were utilized to evaluate thermal comfort. These instruments, including thermometer, hygrometer, Anemometer, Radiant temperature meter, Wearable devices that measures skin temperature, sweat rate, and other physiological factors [101–103] and thermal cameras that visualize and map temperature distribution in a space. The gathered data enables the computing of indices such as Predicted Mean Vote (PMV) and Predicted Percentage of Dissatisfied (PPD) to assess thermal comfort in indoor environments like buildings. With the advancement of technology, IoT and IoT with BIM integration are now utilized to monitor indoor environments [98,104–108].

In recent studies, different approaches have been taken to predict thermal comfort in buildings. [109] utilized an artificial neural network for southern European buildings, while [110] combined BIM and BES software for calculation. [50] developed a BIM and IoT integration that sends automated alerts to building operators and FM through a user interface in a residential apartment in Canada. [52] extended this method to an educational building. These studies use temperature and humidity sensors to detect thermal discomfort, and some have explored the integration of BIM with IoT-based wireless sensor networks for environmental monitoring and emotion detection. However, the current framework is limited as it focuses mainly on temperature, humidity, and occasionally CO2 levels, neglecting air velocity and radiant temperature. This approach, although useful, represents a digital shadow rather than a digital twin.

Qian [111] conducted a year-long survey on Thermal Sensation Votes (TSV) using a digital twin platform to understand environmental parameters in the interior and semi-open spaces and found that thermal discomfort was experienced during summer day and night. Hosamo

[86] integrated real-time sensor data, Bayesian network model, occupants survey, and probabilistic model via the Revit and Dynamo plugin for intelligent detection and prediction of faults causing discomfort in buildings. These studies suggest limitations in current frameworks, focusing primarily on temperature, humidity, and sometimes CO2 and neglecting air velocity and radiant temperature.

Deng [112] used a digital identity in understanding indoor thermal comfort for human-centric monitoring in smart buildings. A group of two scenarios was created for non-digital ID and Digital ID. This shows an improvement in digital ID participants in thermal sensations through feedback from respondents. However, the study only focuses on two parameters of indoor thermal comfort which are relative humidity and air temperature of the room. Furthermore, Abdelrahman et al. [49] adopted Build2Vec where spatial-temporal data was used in predicting the personal thermal comfort of occupants using K-Neural Neighbors (KNN) algorithms which were said to be more accurate than the conventional method. The Build2Vec adopted spatial data from BIM and indoor localization in the real world which is 14–28% more accurate than the conventional way of predicting occupants' satisfaction through statistical and machine learning. Graph neural network (GNN) was used as integration for the BIM model, occupant data (wearable sensors and survey), and environmental data [113]. These four sources were combined in Dynamo in Revit. Cypher files of each were integrated into the graph neural network platform (Neo4J) to draw the relationship between different elements. This was able to understand human thermal comfort and energy saving for room

This innovative source of subjective feedback mechanisms from occupants through smart watches, polling stations, activity workspaces, and QR code-based sustainability stations was adopted by [49]. This study used thirty (30) users to predict their thermal comfort using various technologies such as a smartwatch. Though, this was based on temporary data which makes validity and reliability questionable. Shahinmoghadam et al. [83] went further to adopt VR technologies as interaction with the indoor virtual environment using a game engine. However, this study was conducted in a residential building and one person was engaged in responding to thermal preferences which were biased in representation. The study did not predict the energy, thermal preference, and sensation of various users. Recent applications are stated in Table 5.

5.2. Digital twins and energy consumption in buildings

Energy consumption is a pressing issue globally as urban population growth exacerbates the challenge of meeting growing energy demand. Energy resource depletion is having significant impacts on the environment, including global warming, ozone depletion, and climate change. A study by Medjdoub and Chalal [116] found that primary energy consumption increased by 85% and carbon dioxide emissions rose 75% from 1980 to 2012, with average annual increases of 1.9% and 1.8%, respectively. This highlights the need to address energy consumption and its impact on the environment. In recent years, energy consumption in countries like Russia, India, Brazil, China, and South Africa has risen 4% annually, exceeding that of developed G7 countries.

In the built environment, the building sector accounts for 40% of total energy consumption and is expected to increase with population growth, building age, and increased demand for comfort [6]. The rate of energy consumption depends on the use of lightings, electrical appliances and control of HVAC to suit comfort. According to US Energy Information Administration (2015), energy consumption in the built environment is projected to increase by 26.9% in the next 27 years, with an average rate of 0.9% per year. By 2040, energy consumption in residential and non-residential/commercial buildings is expected to account for 58% and 42% of total energy consumption, respectively.

Efforts have been made to achieve net-zero energy building (NZEB) and reduce energy consumption in other sectors [117]. Retrofitting existing buildings is one such concept, but it faces challenges such as disruptions and costs, and may not meet occupants' comfort and energy-saving preferences [99]. New technologies have emerged as an alternative to reduce energy consumption in the construction industry and built environment. Studies that have explored these technologies for energy monitoring, visualization, and management in buildings will be discussed below.

In the effort to minimize energy consumption, IoT-enabled control mechanisms have been developed and studied. Hossein Motlagh [118] used a light sensor in an IoT platform to monitor indoor light variation, leading to a decrease in power consumption. [119] also developed a scalable data-driven digital twin based on a gaussian process (GP), allowing for accurate prediction and control which is more faster and less expensive than other building models such as Energy Plus (*E*+). Also, other process apart from GP can be used for scalable data driven DT. *E*+ and other simulation tools lack control over physical entities, known as a digital shadow. The *E*+, TRNSYS, and ANSYS are simulation tools that can be classified under the white box model and require high expertise, and time in building such a model. It requires information such as geometry, operation schedule, design, and equipment and materials properties. The major setback of these models are that they cannot be used for advanced control like model predictive control (MPC).

Dave [120] developed a web-based platform integrating IoT sensors with energy consumption and occupant comfort campus data to enhance facility management efficiency through informed decision-making. Vering [121] developed a digital twin prototype that communicates prediction set for different scenarios via MQTT interfaces for increased system efficiency. Machado et al. [122] integrated BIM and IoT for real-time building energy performance monitoring using a visual programming language. Furthermore, Francisco, Truong [47] showed the potential of representing measured energy consumption data in as-built BIM to drive energy-saving behavior and improve automatic transmission of sensor information to BIM platforms.

Kaewunruen et al. [6] found NZEB technologies (e.g. Solar PV with wind turbines) are feasible for existing buildings, with a 23-year return period based on UK case study. Jafari [44] improved building energy performance using BIM and Energy Plus, but with no automatic feedback to physical entities (i.e. digital shadow). Seo and Yun [13] in South Korea found a turn-off strategy reduced energy consumption in a university classroom building by 60% with LED lighting, and revealed the LED is highly designed that consumes more energy which needs adjustment. Nevertheless, the study did not consider cost implications for LED and whether occupant will be satisfied with the lighting performance.

In Italy, Schweigkofler [14] used BIM-IoT integration for energy management in educational buildings. The study employed a flexible, cloud-based solution for real-time indoor environmental monitoring with a focus on energy management. However, it has limitations, such as limited sensors used and absence of automatic feedback and control system in buildings. Other studies integrate building energy simulation or facilities monitoring to evaluate energy consumption. This study only focused on indoor monitoring and didn't assess energy consumption of building facilities. These studies utilized various technologies, like sensors, cloud databases (influx database and MySQL), BIM, IoT devices, and Arduino (sends sensor data to BIM). These studies classified their conceptual framework into 3 to 5 layers. These studies show that research has been conducted on the use of IoT, BIM, and their integration for energy monitoring, visualization, and management of existing buildings for lighting.

In the field of building energy management, various studies have explored the use of digital twins (DTs) for improving thermal comfort and energy efficiency. Hosamo, Hosamo [70] used an artificial neural network and a multi-objective genetic algorithm to optimize thermal comfort and energy consumption, saving 10.8% to 13.2% energy during the summer period. There is need for further studies for other seasons of the year to understand the percentage of energy savings. [56] aimed to reduce energy consumption by 1% through a DT control center for

Table 5
Recent applications of Digital twins in thermal comfort.

Authors	Research Method	Target	Software/Tools	Role/Function	Machine learning method
Hosamo, Nielsen [86]	Case study	Comfort and fault prediction	Sensors, Probabilistic model, Bayesian networks	Understanding comfort, discomfort in humans and fault detection of systems	Bayesian Network
Gnecco, Vittori [113]	Experiment	Human comfort and energy saving	BIM Model, Environmental sensors, wearable sensors, survey	Unified platform for occupant information and environmental sensors	Graph Neural Network
Qian, Leng [111]	Case study	Occupant thermal comfort	Survey, Digital twin platform (Neo4J)	Monitoring of environmental parameters, understanding the comfort and discomfort level of users for a year	Linear regression
Deng, Wang [112]	Case study	Human-centric monitoring	Unity, SQL, COZIR Sensors	Real visualization of temperature and humidity	Random forest
Abdelrahman, Chong [49]	Experiment	Personal thermal comfort of occupants	BIM, Build2Vec	Predicting personal comfort	KNN Algorithm
Abdelrahman and Miller [114]	Case study	Thermal preference of occupants	Smartphones/watches, BIM and GNN (Graph Neural Network)	Optimize thermal comfort subjective data using spatial proximity	N/A
Valinejadshoubi, Moselhi [50]	Experiment	Automated alert to Operator if set points are exceeded	BIM, IoT, Dynamo	Visualization and Monitoring	None
Shahinmoghadam, Natephra [83]	Case study	Thermal comfort conditions	BIM, IoT, VR	Real-time Monitoring	N/A
Nurumova, Ramaji [115]	Case study	Thermal monitoring	Wireless sensors, BIM, Bluetooth (BLE), Cloud-based IoT platform, TwinUp	Monitoring of humidity and temperature	N/A
Zaballos, Briones [52]	Case study	Monitoring of thermal comfort	BIM, IoT, Dynamo, Wise Module, Arduino, SQL	Optimizing comfort	N/A
Amoruso, Dietrich [110]	Experiment	Thermal comfort improvement	BIM	Thermal comfort analysis	None
Escandón, Ascione [109]	Case study	Thermal comfort	BIM, MATLAB	Thermal comfort prediction	ANN

a hospital in China, but the results were not practical and proven. Ni, Eriksson [123] created a cloud-based DT framework for historic public buildings in Sweden, using communication, computing technologies and machine learning algorithms to predict energy consumption. Clausen, Arendt [45] explored DTs for thermal comfort and energy efficiency in smart buildings, but the study did not consider participant opinions on thermal comfort or quantify potential energy savings.

Agostinelli [81] studied cyber-physical systems and AI to improve energy management in Rinascimento III in Rome, using machine learning systems to integrate self-production and supply systems in an energy smart grid. Song [124] proposed a data-model fusion dispatch strategy based on DTs for fault tolerance and privacy protection. Many studies have focused on visualizing, tracking, optimizing, or monitoring thermal comfort and energy, but not on interactive and intuitive use of DT applications. Recent applications are shown in Table 6

6. Discussion, limitations and future directions

6.1. Discussion

The results of this review provide insight into the use of AI and machine learning in optimizing smart buildings, as evidenced by the high coverage and frequency of keywords related to digital twin, energy consumption, energy efficiency, energy utilization, buildings, architectural design, machine learning, smart city, and artificial intelligence. However, the results also indicate a need for further research in areas such as extended reality, mixed reality, personal thermal comfort model, and indoor air quality, which showed lower coverage. This highlights the importance of continued exploration in these areas to fully realize the potential of AI and machine learning in smart building optimization.

The utilization of digital twin technology (DT) for thermal comfort and energy efficiency is a widely researched topic, as indicated by the high number of citations in 21 articles, as shown in Table 3. Despite the growing focus on energy consumption, utilization, and efficiency, there is a need for more attention to be given to the aspect of thermal comfort and occupant preferences. The study showed that there is notable research in selected regions of the world, including the UK, USA, China, Spain, Italy, Singapore, Germany, Australia, Finland, and Canada. However, other countries, such as parts of Africa, South America, the Middle East, and Asia, are yet to fully explore this area, highlighting the need for more international collaboration.

The results of the critical review of digital twin applications for thermal comfort and energy efficiency suggest areas for improvement and future considerations. While much of the focus has been on monitoring and visualizing relative humidity and air temperature, other factors such as air velocity and mean radiant temperature (MRT) have received less attention. The use of thermal imaging cameras to assess skin temperature and personal comfort has been studied, but MRT, a key component in thermal comfort, has not been adequately addressed. Further research is needed to examine the impact of heat gain in buildings on occupant comfort, especially through the use of thermal imaging cameras in comparison to black globe thermometers.

Incorporating the subjective perception of occupants into the technical setup used for monitoring and visualizing thermal comfort parameters has received growing attention in recent years [49,113]. However, few studies have effectively considered physiological responses as a means of understanding the subjective experience of thermal comfort. Shahinmoghadam, Natephra [83] being one of the few studies that have attempted to address this gap by exploring the use of XR technologies for intuitively incorporating personal factors such as clothing insulation and metabolic rate in real-time with environmental parameters. However, more research is needed to fully understand the role of subjective perception in the monitoring and visualization of thermal comfort.

Several tools and technologies have been developed to optimize and increase energy efficiency in existing buildings, including simulation, storage, computing, and prediction. These tools have been applied to various building systems such as HVAC, lighting, and electrical appliances, as highlighted in Table 6. In terms of DT for energy efficiency and optimization, several algorithms have been used, including ANN, DNN, MPC, AI, and GA. These algorithms have shown promise in predicting

Table 6
Recent applications of Digital twins in energy efficiency.

Authors	Research Method	Target	Software/tools	Role/Function	Machine learning method
Song, Xia [124]	Case study	Renewable energy	TRNSYS, MATLAB	Optimization of energy	CNN-LSTM
Tan, Chen [125]	Experiment	Lighting	BIM, video stream, cloud database, VO&M platform	Remote visual management for reduced energy consumption	YOLOv4 algorithms
Seo and Yun [13])	Case study	Lightings	BIM	Energy-saving technologies and strategies for existing building	None
Schweigkofler, Braholli [14]	Case study	Indoor thermal comfort/energy	BIM, IoT, Influx Database, Cloud base service	Visualization	N/A
Henzel, Wróbel [126]	Case study	Electrical Appliances	Influx Database, SQL	Predicting energy consumption	Naive, linear regression, LSTM network, prophet
Hosamo, Hosamo [70]	Case study	HVAC	BIM, Sensors, MATLAB	Energy optimization	Artificial neural network, multi-objective genetic algorithm
Matsuda and Ooka [72]	Case study	HVAC	None	Optimization	Deep neural network
Clausen, Arendt [45]	Experimental	HVAC	SMAP, OccuRE, Zone control application, Archiver, Weather Driver, Xovis Camera Driver, KNX Gateway	Optimization/Controlling	N/A
Desogus, Quaquero [127]	Case study	Thermal comfort/energy	BIM, IoT, Dynamo	Visualization	N/A
Agostinelli, Cumo [81]	Case study	Renewable energy	BIM, Internet of things	Optimization of energy	AI
Fathy, Jaber [128]	Case study	Electrical appliances for household	IoT smart gateway	Visualization	Reinforcement learning
Ni, Eriksson [123]	Case study	Historic buildings	IoT, cloud computing (Microsoft Azure)	Predict, and optimize real-time and historical data	Artificial Intelligence
Jafari, Zaidan [44]	Case study	Thermal behavior and energy use	BIM, Energy Plus	Monitor thermal behavior	N/A
Machado, Dezotti [122]	Case study	Energy	BIM, IoT, Dynamo	Monitor energy consumption	NA
Jain, Nong [119]	Case study	HVAC control	SCADA/BEMS system, *E+/D+* controller, Energy plus OPC bridge, OPC server	Efficient modeling and control of HVAC	Gaussian process
Dave, Buda [120]	Case study	Energy usage, occupant comfort	BIM, IoT sensors and Otaniemi3D	Visualization of energy	NA

and controlling energy consumption in buildings, but more algorithms are needed to compare the level of accuracy to those used in previous studies.

6.2. Limitations and future directions

Research Gap: Despite the development of several models for thermal comfort and occupant behavior in buildings, these models are not easily generalized due to differences in environmental conditions, psychological and physiological characteristics of individuals. There remains a challenge between thermal comfort and energy consumption [99]. The limitation of the study is that the search of keywords was only limited to one database however this was because a high number of documents were found in Scopus compared to other databases such as web of science or google scholar and covered other databases

Proposed Research Directions: Advances in technology have led to the use of digital tools for monitoring, visualizing, and optimizing thermal comfort parameters. There is a need to understand the best tools for the collection and transmission of thermal comfort data from users and energy. Future research can investigate alternative machine learning algorithms, such as SVM and Gaussian process, for predicting energy consumption in intelligent building systems to determine their accuracy.

In addition, further studies can explore the potential of XR technologies for creating immersive experiences and understanding occupant behavior, preferences, and thermal sensations in real-time. The adoption of additional sensors for monitoring indoor environments, such as CO2 and air velocity, can also be investigated for automated alerts in existing buildings.

Finally, future studies should focus on the real-time quantification of thermal comfort indices in both outdoor and indoor environments, potentially using digital tools and programming. This can serve as a more accurate solution to current formulaic methods and enhance effective thermal monitoring (Fig. 9).

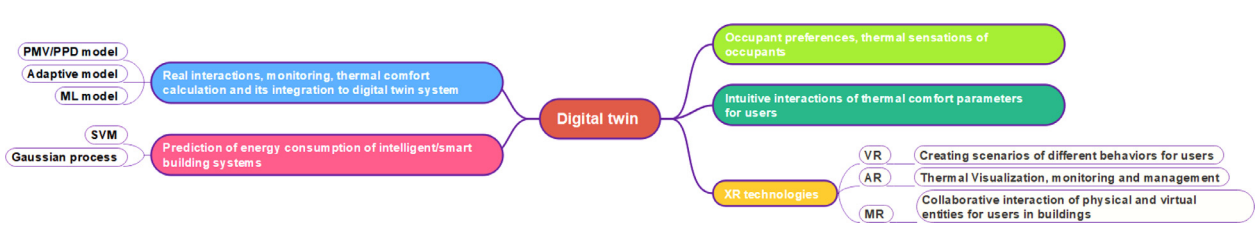

Fig. 9. Future research opportunities.

7. Conclusions

The integration of digital technologies is continuously expanding in the field of thermal comfort and energy for buildings. The DT concept, incorporating multiple digital technologies, has gained increasing attention in recent studies. The review of the current state of the art showed a growing number of studies in this area, with a focus on thermal comfort monitoring, visualization, tracking, energy management, prediction, and optimization for existing buildings. However, there is a need for further international collaboration to expand the studies on digital twins for thermal comfort and energy efficiency. Also, there is a need for further research in areas such as extended reality, indoor air quality, and personal thermal comfort models. The review of the methods, technologies, algorithms, and approaches used in digital twin experiments highlighted the limitations and areas of improvement.

The critical analysis revealed that the adoption of sensors for environmental measures is limited and mostly limited to air temperature and relative humidity. Future research directions include; more sensors for monitoring indoor environments, focusing on real-time quantification of thermal comfort indices, and exploring the potential of XR technologies. The subjective perception of occupants in the building has been examined in a few studies but mostly focuses on monitoring environmental parameters. The research on digital twins for energy consumption showed that monitoring, visualization, optimization, and control at the building and room level have been well-researched. The thermal behavior of buildings has also been studied at different levels, with the optimization of HVAC being crucial for energy efficiency. Predictive algorithms such as reinforcement learning, YOLOv4, ANN, and deep neural networks have been explored by various researchers. There is a need to investigate alternative machine learning algorithms and compare prediction and accuracy levels with others.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: There is no competing financial or personal relationship that can influence the research work

CRediT authorship contribution statement

Victor Adetunji Arowoiya: Conceptualization, Methodology, Investigation, Resources, Writing – original draft, Writing – review & editing. **Robert Christian Moehler:** Resources, Writing – review & editing, Supervision. **Yihai Fang:** Resources, Writing – review & editing, Supervision.

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